UNIVERSAL TECHNOLOGY AND MANAGEMENT UNIVERSITY (UTAMU)

CAI 8103 Capstone Project II — CAT 1: Capstone I Improvement and Capstone II Progress Report

Explainable Spatial Machine Learning for Electricity Distribution-Loss Risk Profiling in Uganda: Leakage-Safe Re-Evaluation and Progress

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Item	Location
Source code	github.com/TBlair24/capstone_electricity — Capstone I in notebooks/; Capstone II in capstone2/ (branch capstone2; also in the ZIP)
Run everything	python capstone2/run_all.py (≈1 minute; 7 pipeline steps then 6 automated tests)
Evidence	outputs/capstone2/tables (23 CSV result files) and outputs/capstone2/figures; console log in evidence/ folder of the ZIP

1. Capstone I Review

Project. The project builds an explainable, spatially enriched machine-learning workflow that profiles electricity distribution-loss risk across Uganda's thirteen distribution licensees (Umeme, eight UEDCL-managed territories and four rural electricity cooperatives). It joins the Electricity Regulatory Authority (ERA) quarterly Distribution Database with open GIS layers (247,704 operational distribution-line features, the transmission network and GADM district boundaries). The outcome is annual loss percentage, 100 × (energy purchased − energy sold) / energy purchased.

Objectives. (1) integrate ERA operational and GIS data at licensee level; (2) describe variation in loss levels and trajectories; (3) form interpretable network archetypes with K-Means; (4) compare tree-based regressors under leave-one-entity-out cross-validation (LOEO-CV); (5) explain the fitted model with SHAP and give cautious decision-support recommendations.

Status at the end of Capstone I. An end-to-end notebook pipeline ran from raw files to 14 figures and 8 processed datasets, and the work was written up as a final report and a journal manuscript draft. Table 1 summarises what was achieved and what remained weak.

Table 1. Capstone I achievements and limitations.

Area	Achieved in Capstone I	Limitation carried into Capstone II
Data integration	107 licensee-years (2009–2022), 13 licensees; 9 predictors after VIF screening	28 of 107 rows had network ratios median-filled before validation (ERA network length blank before ≈2016)
Description	Mean loss 27.8%; NWST 38.5% and BECS 37.7% highest; Umeme declining	Technical/commercial loss fields 68–69% missing, so mechanisms could not be separated
Archetypes	k = 3: rural cooperative, peri-urban UEDCL, urban/dense	Clustered 107 licensee-years, not 13 licensees; no stability test; silhouette only 0.24
Prediction	RF, LightGBM, XGBoost under LOEO-CV; best RMSE 10.76 pp	No model beat a mean baseline ($R^2 \leq -0.006$); trend feature contained the current year's outcome
Explanation	Global and local SHAP plots	SHAP computed on a full-data model that did not generalise
GIS	Line density and transmission-access proxies	Spatial join double-counted transmission lines; maps omitted the country outline

2. Literature Review and Research Gap

The review was extended from seven method references in Capstone I to a critical, thematic synthesis of 30+ studies (full matrix in the manuscript draft). Four themes now drive design decisions.

Customer-level detection does not transfer to licensee-level regulation. Non-technical-loss studies report very high accuracy on smart-meter data — for example Random Forest with oversampling (Abro et al., 2025) and CNN–LSTM networks at about 97% (Elshennawy et al., 2025) — but they classify customers within one utility using random splits. Reviews stress label scarcity, class imbalance and concentration on advanced-metering settings (Glauner et al., 2017; Kgaphola et al., 2024). Uganda's regulator observes 13 licensees annually; these accuracies are not comparable benchmarks.

Population shift is the binding constraint. Parvaz and Ganji (2026) report R^2 0.987 for XGBoost regression of loss severity on one dataset but only 0.125 on a second dataset. Capstone I saw the same effect under LOEO-

CV. Grouped validation is recommended whenever observations are nested in entities (Roberts et al., 2017), and leakage is a leading cause of irreproducible machine-learning results (Kapoor & Narayanan, 2023; Kaufman et al., 2012).

Forecasts must beat a naïve benchmark. The forecasting literature evaluates models against simple reference forecasts — the last observed value — and uses rolling-origin evaluation to test genuine out-of-sample performance (Hyndman & Koehler, 2006; Tashman, 2000). None of the loss-analytics studies above reports such a benchmark, and Capstone I compared only with the pooled mean.

Explanations and clusters need validation. SHAP explains the fitted model, not the physical system (Lundberg & Lee, 2017); robustness testing of explanations is rare in energy applications (Arabzadeh & Frank, 2025), while expert-facing explanations can expose dataset problems (Coma-Puig et al., 2023). Cluster solutions should be checked for bootstrap stability (Hennig, 2007) using the Adjusted Rand Index (Hubert & Arabie, 1985).

Table 2. Critical comparison of key studies and how each shapes Capstone II.

Study	Setting and method	Finding	Weakness	Effect on Capstone II design
Abro et al. (2025)	Customer consumption; 7 classifiers × 7 resamplers	Very high accuracy for RF + oversampling	Random splits, one utility	Not used as a benchmark; entity transfer reported instead
Elshennawy et al. (2025)	SGCC smart-meter data; CNN–LSTM	≈97% accuracy	Dense AMI data unavailable in Uganda	Confirms need for a non-AMI, regulator-level approach
Parvaz & Ganji (2026)	Smart meters; XGBoost regression + SHAP	R^2 0.987 internally, 0.125 externally	External test only secondary	LOEO-CV kept as primary test, with bootstrap CIs
Twesigye (2022)	Kenya, Tanzania, Uganda utilities; case study	Governance shapes utility performance	Qualitative	Explains why structural features alone fail
Coma-Puig et al. (2023)	Production NTL system; XAI	Explanations exposed data shift	Single utility	SHAP moved to held-out rows; imputation flag examined
Hyndman & Koehler (2006); Tashman (2000)	Forecast evaluation methodology	Benchmark against naïve forecasts; rolling origin	Generic	Naïve persistence baseline and rolling-origin test added

Research gap and how it informs the solution. No published study evaluates licensee-level, spatially enriched loss models in a Sub-Saharan African regulatory setting under both entity-transfer and temporal validation against mean and naïve-persistence benchmarks, with leakage controls and out-of-fold explanations. Capstone II therefore (i) separates *structural* risk from *temporal persistence* in three nested feature specifications, (ii) audits and removes target leakage, (iii) reports skill relative to naïve persistence, not only the mean, and (iv) explains only held-out predictions. The aim is unchanged; the evidence standard is higher.

3. Methodological Improvements

Table 3 lists each change and its reason. Every change is implemented in code (Section 4); none is only proposed.

Table 3. Capstone I versus Capstone II methodology.

Component	Capstone I	Capstone II improvement	Why
Trend feature	Rolling 5-quarter loss slope ending in Q4 of the target year	Removed; replaced by loss_lag1 and a slope over years t−3..t−1 only	Old window contained the outcome (leakage)
Missing data	28 rows median-filled before CV	Imputed inside each training fold (sklearn Pipeline); complete-case sensitivity on 74 rows	Prevents test information entering training
Panel	107 rows; specs not comparable	119 licensee-years (2009–2022); all specifications on identical rows	Fair comparison across specifications
Feature design	One mixed predictor set	Structural (9) ⊂ Persistence (+ previous-year loss) ⊂ Forecast (+ lagged slope)	Separates structure from persistence
Baselines	None (R^2 vs pooled mean)	Mean baseline and naïve persistence ($\hat{y}_t = y_{t-1}$)	Tests whether models add skill
Models	RF, LightGBM, XGBoost (tuned on same folds)	Ridge (penalty chosen within fold), RF, XGBoost with fixed hyperparameters	Removes tuning bias; adds a shrinkage model
Validation	LOEO-CV only	LOEO-CV + rolling-origin (2014–2022) + 2,000 licensee-bootstrap CIs + ablation	Unseen licensee and future year; uncertainty
Archetypes	K-Means on 107 years	K-Means on 13 licensee median profiles; 200 bootstrap resamples, ARI	Correct unit; stability evidence

Component	Capstone I	Capstone II improvement	Why
GIS	Spatial join for transmission; no flag	Geometric clip to service areas; gis_imputed flag; coverage table	Removes double counting; transparency
Explainability	SHAP on full-data model	Out-of-fold SHAP (XGBoost) and Ridge coefficient stability across 13 folds	Explains generalising behaviour
Architecture	Single notebook	7 modular scripts, one runner, 6 automated tests, outputs versioned in Git	Reproducible, auditable

4. Implementation Progress

All Capstone II work is in the capstone2/ package (about 870 lines of Python including tests). One command rebuilds every output in about a minute and finishes by running the test suite. Figure 1 is the console output of a clean run: all six tests pass, including a test that perturbs each year's outcome and confirms that its own lagged-slope feature does not change.

```
$ python capstone2/run_all.py
######## gis_features.py ########
districts_total           166.0
districts_assigned         61.0
area_assigned_pct          27.5
operational_dist_km     33606.4
dist_km_assigned_pct       70.0
operational_trans_km     1453.1
 entity  gis_imputed  line_density_gis  trans_km_join  trans_km_clipped  trans_access_join  trans_access_clipped
...
######## build_panel.py ########
Annual rows: 132; analytical panel: 119 licensee-years, 13 licensees, 2009-2022; complete-case rows: 74
...
######## archetypes.py ########
Silhouette (entity level): {2: 0.199, 3: 0.226, 4: 0.237, 5: 0.322, 6: 0.318}
Bootstrap ARI: mean 0.815, median 0.787, share ARI=1: 0.46, share >=0.8: 0.46, 5th pct 0.457
...
######## leakage_audit.py ########
                               Trend feature  RMSE   MAE    R2   Bias  corr_with_current_change
              Capstone I contemporaneous slope 6.833 4.782 0.655 -0.227                     0.742
                    One-quarter-shifted slope 6.313 4.760 0.706 -0.183                     0.816
Strict annual lag, years t-3..t-1 (Capstone II) 9.988 7.447 0.263 -0.031                    -0.220
...
######## tests ########
......                                                                  [100%]
6 passed in 0.35s
Pipeline completed in 66 s
```

Figure 1. Console output of python capstone2/run_all.py from a clean outputs folder (abridged with "..."). Full log in the evidence folder.

4.1 Data and GIS

The panel holds 119 licensee-years from 13 licensees (2009–2022); 74 rows have complete network ratios. Geometric clipping showed that the Capstone I join inflated transmission length heavily: for Umeme from 1,219 km to 406 km (−67%), for NEST 242 to 87 km, and for EST 147 to 12 km (Figure 4b). The coverage audit found that the district lookup assigns 61 of 166 administrative units, covering 27.5% of Uganda's area but 70.0% of mapped distribution-line length; seven licensees needed some GIS imputation and now carry a flag.

4.2 Leakage audit

Table 4. Effect of the trend-feature construction (Persistence specification + trend feature, Ridge, LOEO-CV, same 119 rows).

Trend feature	RMSE (pp)	R^2	Correlation with current-year change
Capstone I contemporaneous slope	6.83	0.655	0.74
Slope shifted back one quarter	6.31	0.706	0.82
Strict annual lag, years t−3..t−1 (adopted)	9.99	0.263	−0.22

Both quarterly constructions correlate strongly with the change in loss they are meant to predict, because their windows overlap the target year; a one-quarter shift still includes Q1–Q3. The interim improved draft's headline R^2 of 0.77 is consistent with this pattern and is superseded. All results below use the strict lag.

4.3 Predictive results

Table 5. LOEO-CV performance on 119 licensee-years (13 folds). 95% CIs from 2,000 licensee-level bootstrap resamples; ΔRMSE < 0 favours the model.

Specification	Model	RMSE	MAE	R^2	ΔRMSE vs mean, 95% CI	ΔRMSE vs persistence, 95% CI
Reference	Mean baseline	11.95	9.33	−0.05	—	0.42 to 4.17
Reference	Naïve persistence	9.67	6.52	0.31	−4.17 to −0.42	—
Structural	Ridge	12.82	10.20	−0.21	−0.04 to 2.34	1.01 to 5.52
Structural	XGBoost	12.75	10.14	−0.20	0.00 to 1.73	1.20 to 5.19
Persistence	Ridge	9.82	7.34	0.29	−3.52 to −0.72	−1.07 to 1.76
Persistence	XGBoost	9.40	7.11	0.35	−3.73 to −1.25	−1.66 to 1.48
Forecast	Ridge	9.99	7.45	0.26	−3.25 to −0.59	−1.03 to 2.02
Forecast	XGBoost	9.57	7.23	0.32	−3.32 to −1.33	−1.55 to 1.56

Random Forest results (Structural 13.12; Persistence 9.68; Forecast 9.63) are in loeo_results_main.csv. RMSE and MAE in percentage points.

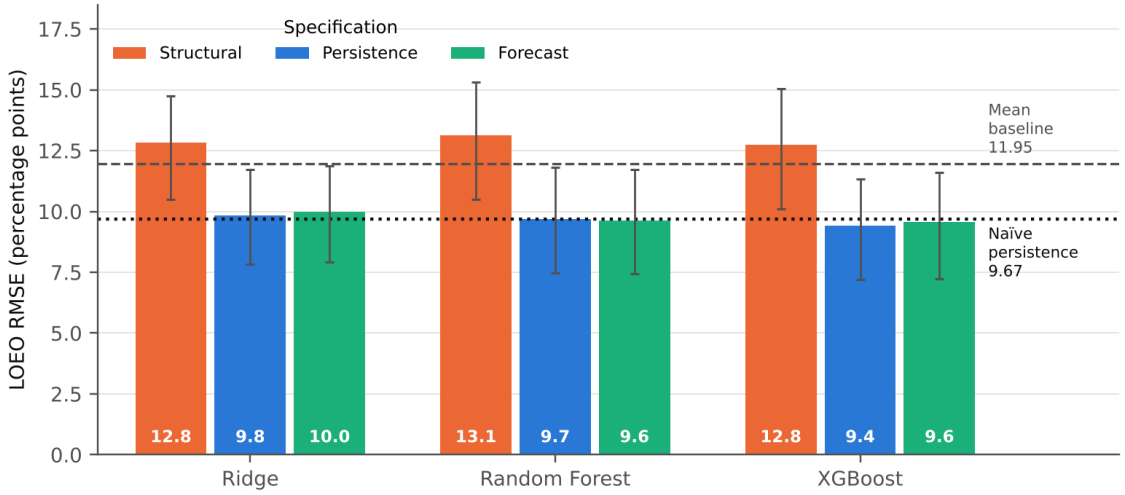


Figure 2. LOEO RMSE by specification and model with 95% licensee-bootstrap intervals. Dashed: mean baseline; dotted: naïve persistence.

Figure 2 and Table 5 show three robust results. First, operational and GIS features alone are no better than the mean (Structural ΔRMSE intervals include or exceed zero), confirming Capstone I. Second, adding previous-year loss gives real, statistically supported skill over the mean (every Persistence/Forecast interval excludes zero). Third, no model beats naïve persistence: all intervals include zero. The Umeme fold shows why: persistence errs by 2.0 pp RMSE, Forecast Ridge by 9.8 pp (bias −7.9), because no other licensee resembles Umeme. In the ablation, lags alone (RMSE 9.27) beat lags plus operational and GIS features (9.99), and the complete-case check (74 rows) gives the same ranking (persistence 7.99; best model 7.89).

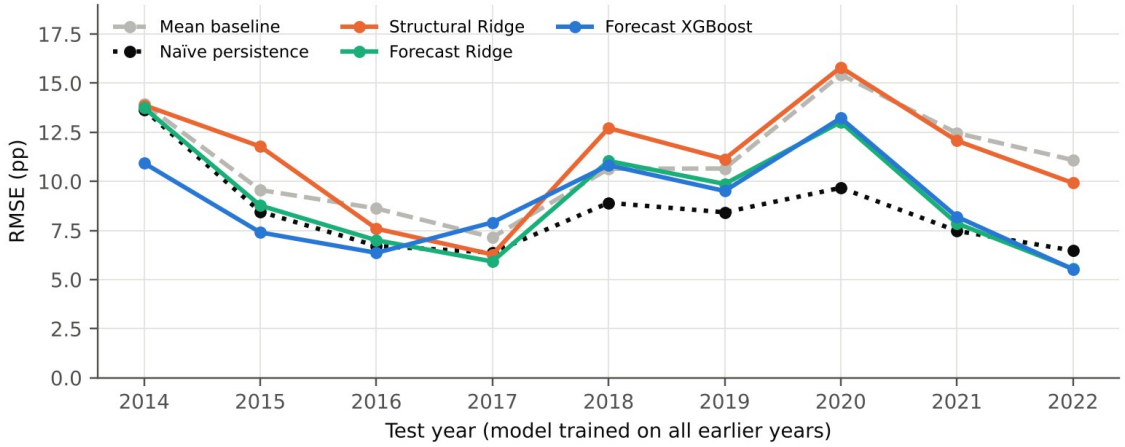


Figure 3. Rolling-origin validation: each model trained on all years before the test year. Pooled RMSE: naïve persistence 8.53, Forecast XGBoost 9.33, Forecast Ridge 9.61, mean 11.46, Structural Ridge 11.74.

Rolling-origin results agree (Figure 3). Models beat persistence in 2014–2016 and 2022 but lose clearly in 2018–2021, worst in 2020 when BECS and NWST losses jumped above 60%. Year-to-year shocks, not structure, dominate the error.

4.4 Archetypes and explanations

At licensee level the three clusters are rural cooperatives (BECS, KIL, KRECS, PACMECS; median loss 21.9%), the eight UEDCL territories (27.9%) and Umeme alone (19.2%). The operational data therefore recover the institutional typology (mapped in Figure 6). Stability is moderate rather than high: mean ARI 0.82, with 46% of resamples reproducing the partition exactly and a 5th percentile of 0.46 (Figure 4a). Silhouette is weak at every k (0.23 at k = 3; maximum 0.32 at k = 5).

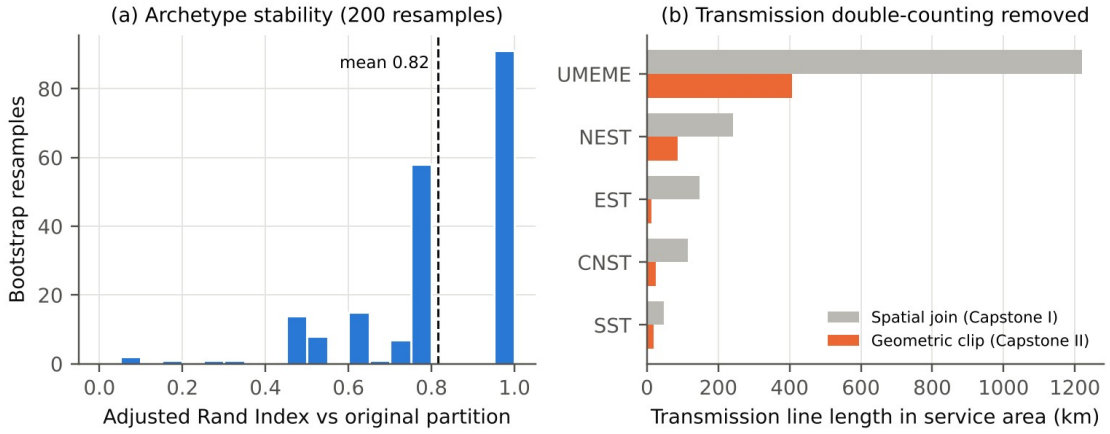


Figure 4. (a) Adjusted Rand Index of 200 within-licensee bootstrap re-clusterings against the original partition. (b) Transmission length per service area: Capstone I spatial join versus Capstone II geometric clip.

Out-of-fold attribution (Figure 5) shows that previous-year loss dominates (mean |SHAP| 4.4 pp), followed by the lagged slope (1.3), O&M per customer (1.1) and staffing (0.9); GIS line density contributes 0.04. Nine of eleven Ridge coefficients keep the same sign in all 13 folds; customer density and transformer density flip. The GIS-imputed flag has a stable positive coefficient: licensees with poorer GIS data also report higher losses, a data-quality signal worth raising with ERA.

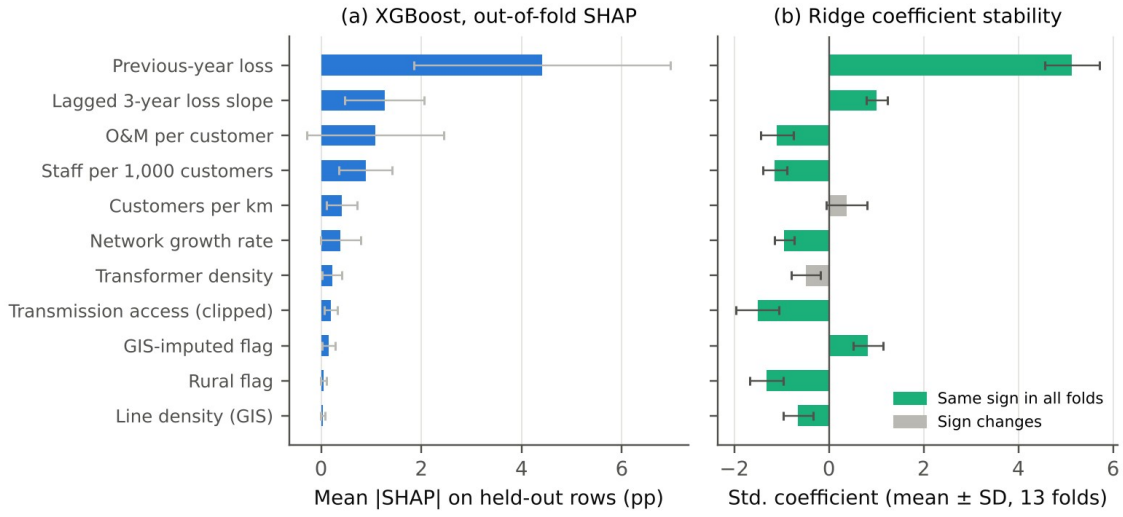


Figure 5. Forecast specification. (a) XGBoost mean |SHAP| on held-out licensee rows (error bars: SD across rows). (b) Ridge standardised coefficients across the 13 LOEO folds (mean ± SD).

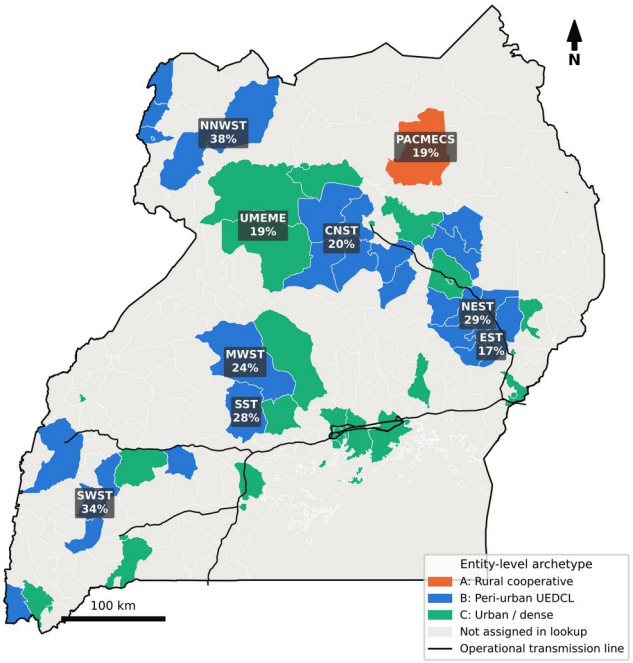


Figure 6. Entity-level archetypes on the full Uganda boundary, with operational transmission lines. Labels: licensee and median annual loss. Grey districts are unassigned; BECS, KIL, KRECS and NWST have no assigned polygons. Not an official concession map.

5. Remaining Work and Completion Plan

5.1 Challenges encountered

- **Leakage invalidated an interim headline.** The strict lag reduced Forecast R^2 from about 0.7 to 0.26; the project narrative now centres on persistence and benchmarking rather than a high-accuracy forecaster.
- **Data gaps.** ERA network length is blank in 40% of quarterly records; technical and commercial loss fields are 68–69% missing; four licensees have only five or six usable years.
- **Spatial coverage.** The lookup assigns 27.5% of Uganda's area; ERA publishes no concession polygons.
- **Structural break.** The Umeme concession ended on 31 March 2025, so post-2025 data will reflect a different operator.

5.2 Outstanding activities and plan

The updated work plan (also supplied as work_plan.xlsx) lists every activity with its status. Dates assume a final examination in early December 2026 and will be shifted if the examination date differs.

Table 6. Updated work plan.

#	Activity	Phase	Status	Target
1	Data ingestion, cleaning, annual aggregation, GIS integration	Capstone I	Completed	—
2	Descriptive analysis, VIF screening, K-Means, LOEO tree models, SHAP, report and manuscript	Capstone I	Completed	—
3	Leakage audit and leakage-safe lag features	Capstone II	Completed	—
4	Clipped GIS features, imputation flag, coverage audit, full-boundary map	Capstone II	Completed	—
5	Three specifications × Ridge/RF/XGBoost, mean and persistence baselines, bootstrap CIs, ablation, complete-case check	Capstone II	Completed	—
6	Rolling-origin validation; entity-level archetypes with bootstrap stability; out-of-fold SHAP	Capstone II	Completed	—
7	Modular pipeline, runner and automated tests; Git branch	Capstone II	Completed	—
8	Ingest ERA Q4 2025 statistics; true prospective test (train ≤ 2022, test 2023–2025), Umeme handover flagged	Capstone II	Outstanding	20 Oct – 2 Nov
9	Model the change from persistence (Δloss target) and report skill relative to naïve persistence	Capstone II	Outstanding	27 Oct – 9 Nov
10	Conformal prediction intervals for licensee-level screening	Capstone II	Outstanding	3 – 9 Nov

#	Activity	Phase	Status	Target
11	Shock analysis 2018–2020 (BECS, NWST) and sensitivity excluding GIS-imputed licensees	Capstone II	Outstanding	3 – 16 Nov
12	Peer-benchmarking screening prototype (archetype z-scores, persistence flags, intervals)	Capstone II	Outstanding	10 – 23 Nov
13	Revise manuscript with clean results; final report; demonstration rehearsal	Capstone II	Outstanding	17 – 30 Nov

Risks and mitigation. If the Q4 2025 file changes its layout, the ingestion step will be adapted and tested before modelling. If prospective results remain at persistence level, the final contribution will be a validated peer-benchmarking and persistence-monitoring tool, which the evidence already supports, rather than a forecaster.

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